Example for bounding off-diagonal terms based on the work of Klurman–Shkredov–Xu

Let α be the Steinhaus random multiplicative function. We shall explain how one shows

$$\mathbb{E}\left|\sum_{n\le x}\alpha(n^2+1)\right|^4 \sim 2x^2.$$

Using orthogonality of α, it is the same as showing that the number of solutions $1 \le a, b, c, d \le x$ to

$$(a^2+1)(b^2+1) = (c^2+1)(d^2+1) \tag{1}$$

that satisfy $(a, b) \notin \{(c, d), (d, c)\}$ is $o(x^2)$; the main term is asymptotic to $2x^2$ arises by taking $(a, b) \in \{(c, d), (d, c)\}$.

We need the following lemma.

Lemma 1 (Divisor bound)**.** *Let $\tau(d)$ be the number of divisors of a positive integer d. Then $\tau(d) = d^{o(1)}$.*

Let Δ be a parameter, to be chosen later. First we count off-diagonal solutions to (1) where at least one of a, b, c, d is at most Δ. By symmetry, suppose $a \le \Delta$. After choosing $a \le \Delta$ and $b \le x$, the product

$$M = (a^2+1)(b^2+1)$$

is fixed, and the number of ordered pairs (c, d) satisfying $(c^2+1)(d^2+1) = M$ is at most $\tau(M)$. Since $M \le (x^2+1)^2$, we have $\tau(M) = x^{o(1)}$ by Lemma 1. Thus the contribution from quadruples (a, b, c, d) with at least one variable at most Δ is

$$O(\Delta x^{1+o(1)}). \tag{2}$$

We now restrict to off-diagonal solutions (a, b, c, d) of (1) with $a, b, c, d > \Delta$. We shall need a variant of Lemma 1.

Lemma 2 (Sum-of-squares bound)**.** *Let $r(d)$ be the number of ways to write d as a sum of two squares. Then $r(d) = d^{o(1)}$. In fact, $r(d) \le 4\tau(d)$.*

Define

$$s := cd - ab, \qquad t := c + d.$$

If $s = 0$ we necessarily obtain diagonal solutions (check!) so we may assume $s \ne 0$. Subtracting the two sides of (1) gives

$$(cd)^2 - (ab)^2 = a^2 + b^2 - c^2 - d^2.$$

Since $cd = ab + s$ and $c^2 + d^2 = t^2 - 2cd$, this becomes

$$s(2ab + s) = (a + b)^2 - t^2 + 2s,$$

or equivalently

$$(a+b)^2 - 2sab = t^2 + s^2 - 2s. \tag{3}$$

Put

$$u := a+b, \qquad v := a-b.$$

Since $ab = (u^2 - v^2)/4$, (3) becomes

$$sv^2 + (2-s)u^2 = 2(t^2 + s^2 - 2s). \tag{4}$$

For fixed s and t, this is a conic in (u, v), except in the special cases $s = 0$ and $s = 2$. The case $s = 0$ is diagonal, as mentioned, and so is irrelevant. If $s = 2$, then (4) gives $v^2 = t^2$, i.e. $a - b = \pm t$; combining this with $cd = ab + 2$ and $c + d = t$ implies that $c(t - c) = a(a \mp t) + 2$ (and that b, d are determined by a, c, t), i.e. $(2a \mp t)^2 + (2c - t)^2 = 2t^2 - 8$. By Lemma 2, there are $\leq x^{o(1)}$ such pairs (a, c).

Whenever $s \notin \{0, 2\}$, we shall apply the following lemma.

Lemma 3 (Integral points on conics)**.** *Let A, B, C be nonzero integers with $|A|, |B|, |C| \leq x^5$. The number of integer solutions $(u, v) \in [-x, x] \times [-x, x]$ to the equation*

$$Au^2 + Bv^2 = C$$

is $x^{o(1)}$.

It implies

$$\#\{(a, b) : a, b \leq x, (4) \text{ holds for fixed } s, t\} \leq x^{o(1)}. \tag{5}$$

Once a, b, s, t are fixed, the pair (c, d) is determined up to order as the two roots of

$$Z^2 - tZ + (ab + s) = 0,$$

so there are at most two choices for (c, d). It remains to count the possible values of s and t. Since all variables are larger than Δ, we have $cd + ab \gg \Delta^2$. Also,

$$|s|(cd + ab) = |(cd)^2 - (ab)^2| = |a^2 + b^2 - c^2 - d^2| \leq 2x^2$$

by (1). Therefore

$$|s| \ll \frac{x^2}{\Delta^2}. \tag{6}$$

There are $O(x)$ choices for $t = c + d$. Combining (5) and (6), the number of off-diagonal solutions with all variables larger than Δ is

$$O\left(x^{1+o(1)}\right) + O\left(x \cdot \frac{x^2}{\Delta^2} \cdot x^{o(1)}\right). \tag{7}$$

Both (2) and (7) are $o(x^2)$ if we choose Δ in the range $x^{1/2+\varepsilon} \leq \Delta \leq x^{1-\varepsilon}$.